Kiran Vaddi

kiranvad.github.io kiran.vaddi@nist.gov
linkedin.com/in/kiranvaddi27 github.com/kiranvad Google Scholar
100 Bureau Dr, Gaithersburg, MD 20899

Education

June 2021 – May 2025
University of Washington, Seattle, WA, USA
Post Doctoral Training in Chemical Engineering
Functional Space Data Representations for Autonomously Engineering Soft Materials

May 2021 – August 2017
University at Buffalo, The State University of New York, Buffalo, NY, USA
Ph.D. in Materials Science and Engineering
Thesis : Representations for data-driven material discovery

June 2016 – June 2012
Indian Institute of Technology Madras, Chennai, TN, India
Dual Degree in Mechanical Engineering
Minor : Industrial Engineering
Thesis : Luminescent solar concentrators using high-contrast gratings

Professional Experience

June 2025 – Present
National Institute of Standards and Technology
Polymers Processing Group
- Develop AI-agents for sustainable and novel biopharmaceutical formulations using the Autonomous Formulation Laboratory.

June 2021 – May 2025
University of Washington, Seattle, WA
Data Science Postdoctoral Fellow, Department of Chemical Engineering
- Developed closed-loop material acceleration platforms emphasizing learning faithful data representations of spectroscopy, and scattering characterizations of polymer, colloid, and soft-matter systems.

June 2017 – June 2021
University at Buffalo, Buffalo, NY, USA
Graduate Research Assistant
- Developed physics-based models for cyclic voltammetry, phase diagrams of polymer blends, and data-driven models for structure-property relations for catalysis and organic photovoltaics.

June 2016 – June 2017
Industrial Interships
- **Caterpillar India Pvt.Ltd :** Finite element-based modeling of thermal stresses in automotive engine components (gaskets)
- **Continental Automotive Components India Pvt.Ltd :** Finite element-based dynamic (vibration and failure) analysis of automotive engine components

June 2016 – June 2017
Indian Institute of Technology Madras, Chennai, TN, India
Undergradte Research
- Construction of constitutive models for diffusion-induced deformation of photopolymer under selective irradiation

Teaching Experience

January 2023 – March 2023
Primary Instructor, Special Topics in Chemical Engineering : Design of Experiments
CHEM E 599 B, Winter quarter, **Evaluation 4.2/5**
- Designed and delivered a graduate-level course on the design of experiments with applications in materials science and chemical engineering.

March 2022 – June 2022
CAPSTONE DIRECT project
CHEME / CHEM / MSE 547 : Molecular Data Science Capstone
- Mentored five graduate students from the Department of Materials Science toward their final project titled "The infinite space of spectroscopy measurements" at the University of Washington.

Publications & Pre-prints

Total 14 publications, 126 citations, h-index 6, i10-index 3, ORCID : 0000-0003-2998-769X

Vaddi, K., Grey, A., Chiang, H. T., Wylie, Z. R., Pozzo, L. D *"Autonomous Phase Mapping of Gold Nanoparticles Synthesis with Differentiable Models of Spectral Shape"*
npj Computational Materials 11, 335, 2025 (Corresponding Author)

HT Chiang, Z Zhang, **K Vaddi**, A Tezcan, L Pozzo *"Efficient Analysis of Small-Angle Scattering Curves for Large Biomolecular Assemblies Using Monte Carlo Methods"*
Applied Crystallography 58 (3), 2025

K Li, **K Vaddi**, S Seifert, J Mata, LD Pozzo *"Self-assembly of a Triblock Copolymer in the Presence of a Rigid Conjugated Polyelectrolyte"*
Macromolecules 57.24 (2024) : 11717-11726.

Chiang, Huat Thart, **Kiran Vaddi**, and Lilo D. Pozzo. *"Data-Driven Exploration of Silver Nanoplate Formation in Multidimensional Chemical Design Spaces"*
Digital Discovery, 2024, 3 (11), 2252-2264

Vaddi, Kiran, Karen Li, and Lilo D. Pozzo. *"Metric geometry tools for automatic structure phase map generation."*
Digital Discovery, 2023, 2 (5), 1471-1483 (Corresponding Author)

Politi, M., Baum, F., **Vaddi, K.**, Vasquez, J., Antonio, E., Bishop, B. P., ..., and Pozzo, L. D. *"High-Throughput Workflow for the Synthesis of CdSe Nanocrystals Using a Sonochemical Materials Acceleration Platform."*
Digital Discovery, 2023, 2, 1042-1057 (Editor's Choice 2023)

Vaddi, K., Liu, H., Pokuri, B. S. S., Ganapathysubramanian, B., and Wodo, O. (2023) *"Construction and high-throughput exploration of phase diagrams of multi-component organic blends"*
Computational Materials Science 216 (2023) : 111829.

Vaddi, Kiran, Huat Thart Chiang, Lilo D. Pozzo *"Autonomous retrosynthesis of gold nanoparticles via spectral shape matching"*
Digital Discovery 1.4 (2022) : 502-510 (Editor's Choice 2022, Corresponding Author)

Lachowski, K. J., **Vaddi, K.**, Naser, N. Y., Baneyx, F., Pozzo, L. D. *"Multivariate Analysis of Peptide-Driven Nucleation and Growth of Au Nanoparticles"*
Digital Discovery 1.4 (2022) : 427-439

Vaddi, Kiran, Olga Wodo *"Active knowledge extraction from cyclic voltammetry"*
Energies 15.13 (2022) : 4575

Elikkottil, A., **Vaddi, K.**, Reddy, K. S., Pesala, B. *"Reduction of Escape Cone Losses in Luminescent Solar Concentrators Using High-Contrast Gratings."*
In Advances in Energy Research, Vol. 1 (pp. 37-43). Springer, Singapore, 2020

Vaddi, Kiran, and Olga Wodo. *"Metric Learning for High-Throughput Combinatorial Data Sets."*
ACS Combinatorial Science 21.11 (2019) : 726-735 (Corresponding Author)

Grant Proposals

August 2025 – July 2027 | **Theory-integrated Autonomous Experimentation for Sustainable Biotherapuetic Formulation Design**
U.S. Department of Commerce, NIST Materials Measurement Laboratory (Awarded FAIN-70NANB25H090-0, **$250k**)
- PI on the grant to develop physics-informed AI-driven agents for desinging biotherapeutic drug formulation by optimizing measurement campaigns (100%).

June 2022 – June 2025 | **AI-Accelerated Optimization of Self-Assembled Organic Mixed Ionic-Electronic Conductors (OMIEC)**
DOE- Basic Energy Sciences-Neutron Scattering (Awarded DE-SC0019911)
- Contributed to this PI-led Energy grant on building AI-driven workflows understanding a novel class of OMIEC materials based on conjugated polymer blends from the Department of Energy (30%).

Awards

February 2025	**Future Investigator Travel Award** American Physical Society Division of Soft Materials
September 2024	**Outstanding Reviewer Award** Royal Society of Chemistry Digital Discovery Journal
December 2023	**Best Poster Award** Materials Research Society Fall Meeting-Symposium on Data Science
March 2023	**Data Science Conference Travel Award** eScience Institute, University of Washington, Seattle, WA
June 2021	**UW Data Science Postdoctoral Fellowship** eScience Institute, University of Washington, Seattle, WA
August 2013	**Outstanding Undergraduate Service Award** National Service Scheme, Indian Institute of Technology Madras

Invited talks

April 2025 Invited talk	**Lila Sciences, Boston, MA, USA** Differentiable Phase Mapping of Nanoscale Materials **Kiran Vaddi**
March 2025 Invited talk	**Colorado State University, Fort Collins, CO, USA** Functional Data Analysis Tools for Accelerated Materials Design **Kiran Vaddi**
December 2023 Invited Panelist	**The Kavli Foundation Frontiers of Materials, Boston, MA, USA** Symposium X - How to Build a Self-Driving Lab Keith A. Brown, John Dunlap, Robert Epps, Jason Hattrick-Simpers, **Kiran Vaddi**
December 2023 Invited talk	**Materials Research Society, Boston, MA, USA** A Tutorial on Functional Data Analysis for High-Throughput Experiments **Kiran Vaddi**
October 2023 Invited talk	**UW Data Science Seminar, Seattle, WA, USA** Functional data analysis tools for autonomous experimentation **Kiran Vaddi**
August 2023 Invited talk	**American Chemical Society Fall Meeting, San Francisco, CA, USA** Cold, warm, warmer, hot! Impact of distance metrics on autonomous experimentation **Kiran Vaddi**, Kacper Lachowski, Huat T Chiang, Karen Li, Lilo D. Pozzo
January 2020 Invited talk	**IEEE Rochester Section Chapter, Rochester, NY, USA** Function Space Data Representation of Temporal Signals for Machine Learning **Kiran Vaddi**

Select Conference Presentations & Posters

March 2025 Contributed talk	**American Physical Society March Meeting, Anaheim, CA, USA** Autonomous Phase Mapping of Gold Nanoparticles Synthesis with Differentiable Models of Spectral Shape **Kiran Vaddi**, Huat T Chiang, Lilo D. Pozzo
August 2024 Contributed talk	**American Chemical Society Fall Meeting, Denver, CO, USA** Understanding Design Rules of Colloidal Self-Assembly Using Autonomous Phase Mapping **Kiran Vaddi**, Huat T Chiang, Lilo D. Pozzo

April 2024 | **Materials Research Society Spring Meeting, Seattle, WA, USA**
Contributed talk | High-Throughput Structural Investigation of Block Copolymer and Conjugated Polymer Co-Assemblies
Kiran Vaddi, Karen Li, Lilo D. Pozzo

December 2023 | **Materials Research Society Fall Meeting, Boston, MA, USA**
Poster | Function Space Representations for Complex Material Workflows
Kiran Vaddi (Best poster award)

March 2023 | **American Physics Society March Meeting, Las Vegas, NV, USA**
Contributed talk | Metric geometry tools for automatic structure phase map generation
Kiran Vaddi, Karen Li, Lilo D. Pozzo

August 2022 | **Gordon Research Conference on Computational Materials Sciences, Newry, ME, USA**
Poster | Shape-based data representations for experimental spectra
Kiran Vaddi, Huat T Chiang, Karen Li, Lilo D. Pozzo

May 2022 | **Materials Research Society Spring Meeting, Honolulu, Hawaii, USA**
Contributed talk | Autonomous retrosynthesis of nanoscale structures via spectral shape matching
Kiran Vaddi, Huat Thart Chiang, Lilo D. Pozzo

December 2020 | **Materials Research Society Fall Meeting, Boston, MA, USA**
Contributed talk | High-throughput exploration of materials-phase diagram maps in multi-component organic blends
Kiran Vaddi, Balaji Pokuri, Baskar Ganapathysubramanian, Olga Wodo

October 2020 | **AI for Materials : From Discovery to Production, Webinar,**
Poster | Probabilistic representation of cyclic voltammetry curves for data-driven material discovery
Kiran Vaddi, Olga Wodo

December 2019 | **Materials Research Society Fall Meeting, Boston, MA, USA**
Contributed talk | Accelerating catalyst discovery using Gaussian processes and active learning
Kiran Vaddi, Olga Wodo, Krishna Rajan

May 2019 | **Toyota Research Institute Accelerated Material Design and Discovery Meeting, Boston, MA, USA**
Poster | Machine Learning-Based Simulation Tools for Combinatorial Experiments
Kiran Vaddi, Olga Wodo, Krishna Rajan

Research mentoring

June 2021 – Present | **Graduate student mentoring, Pozzo Research Group, University of Washington, Seattle, WA**
- Kacper Lachowski, Ph.D.
- Huat That Chiang, Ph.D. candidate
- Karen Li, Ph.D. candidate

September 2024 – Present | **Undergraduate student mentoring, Pozzo Research Group, University of Washington, Seattle, WA**
- Aleks Grey, Chemical Engineering.

December 2019 | **Partnerships for Research and Education in Materials (PREM), MRS, 2019, Boston, USA**
- Mentored two undergraduate material science students during their visit to the MRS 2019 Fall meeting.
- Advised mentees towards successful abstract writing, poster competitions, and networking sessions.

Service, Outreach & Leadership

June 2021 – Present | **Academic service**
- **Peer review :** RSC Digital Discovery (**Outstanding Reviewer 2023**), Neural Information Processing Systems (NeurIPS), Journal of Electroanalytical Chemistry, Matter, Nature communications, RSC Advances, npj Computational Materials, Scientific Reports.
- **Committees :** UW ChemE Distinguished Young Research Scholar Seminar reviewer (2022, 2024), Postdoc representative of 2023 faculty hiring committee.

June 2021 – December 2022 | **Co-chair eScience Postdoc Seminar, University of Washington, Seattle, WA**
- Organized a year-long seminar series for the eScience Institute at the University of Washington, Seattle including invited talks about research and career development.

2020 | **Open source software**

github.com/kiranvad

- **AFL-agent** : AI agent for the Autonomous Formulation Laboratory.
- **pyMECSim** : Simulating voltammograms of complex multi-step electrochemical reactions in Python.
- **geomstats** : Performing geometric data analysis on functional data.
- **polyphase** : Computing phase diagrams of polymer blends using Flory-Huggins theory.

June 2012 – June 2015 | **National Social Service Scheme, IIT Madras, Chennai, India**

- Awarded **Outstanding Undergraduate Service Award** in 2013 for organizing and mentoring the 'IIT Madras for The Banyan' project.
- Co-organized the inaugural, student-led, interactive support sessions for persons living through poverty and homelessness via The Banyan
- Developed teaching materials and actively participated in community teaching programs.

Professional Memberships

2018 Materials Research Society
2023 Americal Physical Society (DPOLY, DSOFT, GDS)
2023 American Institute of Chemical Engineers (CoMSEF, MESD)
2023 American Chemical Society (COMP, PMSE)